

Program 9: Organize Kubernetes Pods Using Labels and Namespaces

Step 1: Start the minikube and check status

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~$ cd p9
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ minikube start
🐹 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚀 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
  ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ▪ Using image registry.k8s.io/ingress-nginx/controller:v1.13.2
  ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
  ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
🔍 Verifying ingress addon...
🌞 Enabled addons: default-storageclass, storage-provisioner, metrics-server, ingress

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
  ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check status of minikube

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create Namespaces (namespaces.yaml)

```
GNU nano 7.2
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 4: Create pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step 5: Apply those

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl apply -f pods.yml
pod/frontend-pod created
pod/frontend-pod created
pod/backend-pod created
pod/backend-pod created
```

Step 6 : Get the pods created

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f966444b-5m82f    1/1     Running   1 (6d22h ago)  7d
nginx-deployment-6f966444b-krd2k    1/1     Running   1 (6d22h ago)  7d
nodejs-pod                          1/1     Running   1 (2m59s ago)  7d
test-cron-29435296-4hgx9            0/1     Completed 0              2m39s
test-cron-29435297-4d75s           0/1     Completed 0              2m6s
test-cron-29435298-frgsk            0/1     Completed 0              66s
test-cron-29435299-c6dh9           0/1     ContainerCreating 0              6s
```

Step 7 : Organised them based on namespaces and labels

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl get pods -n dev
NAME            READY   STATUS    RESTARTS   AGE
backend-pod     1/1     Running   0           16s
frontend-pod    1/1     Running   0           17s
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl get pods -n prod
NAME            READY   STATUS    RESTARTS   AGE
backend-pod     1/1     Running   0           19s
frontend-pod    1/1     Running   0           19s
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl get pods -n dev -l app=frontend
NAME            READY   STATUS    RESTARTS   AGE
frontend-pod    1/1     Running   0           32s
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl get pods -n dev -l app=backend
NAME            READY   STATUS    RESTARTS   AGE
backend-pod     1/1     Running   0           35s
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl get pods -n prod -l app=backend
NAME            READY   STATUS    RESTARTS   AGE
backend-pod     1/1     Running   0           41s
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p9$ kubectl get pods -n prod -l app=frontend
NAME            READY   STATUS    RESTARTS   AGE
frontend-pod    1/1     Running   0           47s
```